

Multiple Choice

1. **Answer:** C

Options: A) $P(H, U, P, W) = P(H) * P(W) * P(P) * P(U)$; B) $P(H, U, P, W) = P(H) * P(W) * P(P | W) * P(W | H, P)$; C) $P(H, U, P, W) = P(H) * P(W) * P(P | W) * P(U | H, P)$; D) None of the above

Note: Correct option: C - $P(H, U, P, W) = P(H) * P(W) * P(P | W) * P(U | H, P)$

2. **Answer:** C

Options: A) Is unbounded, encompassing all real numbers.; B) Is unbounded, encompassing all integers.; C) Is bounded between 0 and 1.; D) Is bounded between -1 and 1.

Note: Correct option: C - Is bounded between 0 and 1.

3. **Answer:** A

Options: A) Decreases model bias; B) Decreases estimation bias; C) Decreases variance; D) Doesn't affect bias and variance

Note: Correct option: A - Decreases model bias

4. **Answer:** D

Options: A) It relates inputs to outputs.; B) It is used for prediction.; C) It may be used for interpretation.; D) It discovers causal relationships

Note: Correct option: D - It discovers causal relationships

5. **Answer:** A

Options: A) random crop and horizontal flip; B) random crop and vertical flip; C) posterization; D) dithering

Note: Correct option: A - random crop and horizontal flip

True / False

6. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

7. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'True, True'

8. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'convolutional networks'

9. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'False, True'.

10. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

Long Form Response

11. **Answer:** `def text_match_one(text): | return 'Found a match!'`

Note: Students should provide a detailed explanation referencing algorithm steps.

12. **Answer:** `def count_range_in_list(li, min, max): | return ctr`

Note: Students should provide a detailed explanation referencing algorithm steps.

End of Answer Key